Student Rollout
$x \sim \mathrm{D}$
π_θ
Student
x
y_1
$\cdots$
y_{t-1}
y_t
$s_t = (x, y_{<t})$
Student Policy Gradient
$\nabla_{z_t} \log \pi_\theta(y_t \mid s_t)$
✓ A_t ✗
×
$\mathbf{g}_t = A_t \nabla_{\mathbf{z}_t} \log \pi_\theta \left(y_t \mid s_t\right)$
Teacher Policy Shift
$\pi_\mathrm{T}(\cdot \mid s_t)$
Teacher
$\pi_\mathrm{T}^\mathrm{ref}(\cdot \mid s_t)$
Teacher Reference
−
$\Delta_t = C\left(z_\mathrm{T} - z_\mathrm{T}^\mathrm{ref}\right), \quad \mathbf{d}_t = \Delta_t / \|\Delta_t\|_2$
Directional Gradient Scaling
$\tilde{\mathbf{g}}_t = \mathbf{g}_t + \lambda_t \, \mathrm{Proj}_{\mathbf{d}_t}(\mathbf{g}_t)$
$= (1+\lambda_t)\, \mathrm{Proj}_{\mathbf{d}_t}(\mathbf{g}_t) + \mathbf{g}_t^\perp$
Project
$\mathrm{Proj}_{\mathbf{d}_t}(\mathbf{g}_t) = u_t \, \mathbf{d}_t,$
$u_t = \left\langle \mathbf{d}_t, \mathbf{g}_t \right\rangle$
$\mathbf{g}_t$
θ_t
$\mathbf{d}_t$
Amplify
$\tilde{\mathbf{g}}_t$
$\mathbf{g}_t^\perp$
$(1+\lambda_t)\, \mathrm{Proj}_{\mathbf{d}_t}(\mathbf{g}_t)$
Learning Beyond Teacher
GRPO
OPD
OPRD
$\mathbf{g}_t$
π^*
π_T
$\tilde{\mathbf{g}}_t$
$\mathbf{d}_t$
π_θ^0
$\pi_\mathrm{T}^\mathrm{ref}$

OPRD amplifies the component of $\mathbf{g}_t$ along $\mathbf{d}_t$ instead of targeting π_T, preserving the stationary points (i.e., $\tilde{\mathbf{g}}_t = \mathbf{0} \iff \mathbf{g}_t = \mathbf{0}$).